Summary Report: Data Cleaning and Feature Engineering Pipeline

1. Project Overview

This project focuses on building a complete data preprocessing and feature engineering pipeline for customer purchase analysis. The objective is to transform raw, unstructured data from multiple sources into a clean, structured dataset suitable for machine learning applications, specifically for predicting customer purchase behavior.

2. Techniques Used

The following data preprocessing and transformation techniques were applied:

- **Data Cleaning:**
  Missing values were handled using multiple techniques including Simple Imputer (mean and mode), Random Sample Imputation, KNN Imputer, and MICE (Iterative Imputer). Complete case analysis was also performed for comparison.
- **Outlier Handling:**
  Outliers were detected and treated using Z-score, Interquartile Range (IQR), and Percentile methods. Winsorization was additionally used to cap extreme values.
- **Exploratory Data Analysis (EDA):**
  Univariate, bivariate, and multivariate analyses were performed using histograms, boxplots, correlation heatmaps, and pairplots to understand data distribution and relationships.
- **Feature Engineering:**
  New features such as purchase_per_day and days_since_last_purchase were created. Log and power transformations were applied to reduce skewness. Binning and binary feature creation were also implemented.
- **Encoding & Scaling:**
  Categorical variables were encoded using Label Encoding, One-Hot Encoding, and Ordinal Encoding. Numerical features were scaled using StandardScaler, MinMaxScaler, and RobustScaler.

3. Key Issues in Raw Data

The initial dataset contained several challenges:

- Missing values in both numerical and categorical columns
- Presence of extreme outliers in income and transaction amount
- Skewed distributions in numerical features
- Mixed-format variables (IDs containing both text and numbers)

- Lack of meaningful derived features

4. Best Techniques Selected

After comparing different methods:

- **Imputation:**
  MICE was most effective for numerical data, while Most Frequent Imputation worked best for categorical data.
- **Outlier Handling:**
  IQR and Winsorization were found to be more robust for skewed data.
- **Scaling:**
  StandardScaler and RobustScaler provided better normalization depending on feature distribution.

5. Final Outcome

A fully cleaned and transformed dataset was generated and exported as **processed_customer_data.csv**. The final dataset contains no missing values, reduced impact of outliers, and includes meaningful engineered features. This dataset is now suitable for training machine learning models for customer purchase prediction.

6. Conclusion

The project successfully demonstrates a complete data preprocessing pipeline, covering data cleaning, transformation, and feature engineering. The final dataset is structured, consistent, and optimized for predictive modeling, ensuring better performance in future machine learning tasks.